

# Midterm Exam

Units 1–11 · pages 7–163 · 100 points · 90 minutes · Closed book

Name \_\_\_\_\_

Date \_\_\_\_\_

Score / 100

*This exam covers the entire first half of the course, with additional weight on Units 8–11. Circle one answer for multiple choice. Write on the lines provided; use the back of the page if you need more room. Budget roughly 40 minutes for Parts A and B, 25 minutes for Part C, and 25 minutes for Part D.*

## Part A · Multiple Choice

25 questions × 2 points = 50 points

1. In the geographic methodology described in Unit 1, what step immediately follows asking a geographic question?
  - a) Explore geographic data
  - b) Publish the results
  - c) Analyze geographic information
  - d) Acquire geographic resources
2. A GPS receiver finds its distance from a satellite using  $D = rt$ . In that equation,  $r$  is
  - a) the rate of Earth's rotation
  - b) the velocity of the receiver
  - c) the radius of the Earth
  - d) the speed of light
3. The first neutral atoms, the ones that included electrons, formed about how long after the Big Bang?
  - a) A few seconds
  - b) A few minutes
  - c) 380,000 years
  - d) 1 billion years
4. A star remains on the main sequence for as long as it can
  - a) continue expanding outward
  - b) fuse elements heavier than iron
  - c) avoid collecting interstellar dust
  - d) balance the inward force of gravity against the outward force of nuclear fusion
5. In the analogy that compresses all of geologic time into one year, the origin of the solar system at 4.57 Ga falls on
  - a) January 1st
  - b) March 1st
  - c) November 16th
  - d) December 31st

6. The boundary between the Cretaceous and the Paleogene coincides exactly with
  - a) the beginning of the Pleistocene ice age
  - b) the Cambrian Explosion
  - c) the extinction of the non-avian dinosaurs
  - d) the first animals crawling onto land
7. Which statement correctly distinguishes a map's extent from its resolution?
  - a) Extent is the area visible on the map; resolution is the smallest unit mapped
  - b) Extent is the smallest unit mapped; resolution is the area visible
  - c) Both describe how far the map is zoomed in
  - d) Extent applies only to paper maps; resolution only to digital ones
8. The Robinson projection is an example of a
  - a) conformal projection
  - b) equal-area projection
  - c) compromise projection
  - d) interrupted projection
9. Of the solar energy reaching the outer atmosphere, roughly what share is in ultraviolet wavelengths?
  - a) About 7%
  - b) About 20%
  - c) About 44%
  - d) About 60%
10. In the Northern Hemisphere, the autumnal equinox occurs on or around
  - a) March 21st
  - b) June 21st
  - c) September 22nd
  - d) December 21st
11. About 99.9% of the mass of the atmosphere lies below which layer?
  - a) The troposphere
  - b) The stratosphere
  - c) The mesosphere
  - d) The thermosphere
12. Santa Ana winds form when the Great Basin east of the Sierra Nevada cools, creating
  - a) a temperature inversion
  - b) a monsoon circulation
  - c) a low-pressure zone
  - d) a high-pressure zone

13. In atmospheric science, *advection* refers to
- a) the horizontal flow of air
  - b) the reflection of light from a surface
  - c) the transfer of energy by electromagnetic waves
  - d) the vertical rising of warm air
14. Nor'easters are
- a) seasonal monsoon winds
  - b) tropical cyclones that strike the Gulf Coast
  - c) mid-latitude cyclones that strike the mid-Atlantic and New England states
  - d) lines of severe thunderstorms along a cold front
15. On the Saffir-Simpson scale, a Category 5 hurricane has sustained winds of
- a) 74–95 mph
  - b) 111–129 mph
  - c) 130–156 mph
  - d) 157 mph and above
16. An *anion* is
- a) an atom with no electrical charge
  - b) a molecule made of two or more elements
  - c) an atom that has lost one or more electrons
  - d) an atom that has gained one or more electrons
17. Halite breaks between layers of sodium and chlorine to form cubes with smooth surfaces. This is an example of
- a) cleavage
  - b) streak
  - c) luster
  - d) fracture
18. When magma breaks off pieces of the surrounding country rock and they fall into the melt, the resulting fragments are called
- a) plutons
  - b) xenoliths
  - c) batholiths
  - d) clasts
19. Obsidian forms when
- a) lava cools so rapidly that crystals do not develop at all
  - b) magma cools slowly over millions of years at depth
  - c) gas bubbles are trapped in cooling lava
  - d) sediment is compacted and cemented

20. Typical *felsic* igneous rocks are
- a) gabbro and basalt
  - b) granite and rhyolite
  - c) peridotite and dunite
  - d) diorite and andesite
21. A mudrock that shows clear evidence of bedding or fine laminations is called
- a) mudstone
  - b) shale
  - c) claystone
  - d) conglomerate
22. All of the dolomite found in ancient rocks formed through
- a) direct precipitation from open seawater
  - b) magnesium replacing some of the calcium in calcite
  - c) the accumulation of marine organism shells
  - d) volcanic activity on the seafloor
23. When clasts in a stream are *imbricated*, they are
- a) sorted perfectly by size
  - b) tilted in the same direction, with their upper ends pointing downstream
  - c) cemented together by calcite
  - d) arranged in vertical stacks
24. A rock buried deeply enough to partially melt, producing a mixture of metamorphic and igneous material, is called
- a) migmatite
  - b) gneiss
  - c) hornfels
  - d) eclogite
25. Greenstone and greenschist form at oceanic spreading ridges by a process the text calls
- a) contact metamorphism
  - b) retrograde metamorphism
  - c) regional metamorphism
  - d) shock metamorphism

**Part B · Matching**

2 questions × 8 points = 16 points

**26. Set 1 — Units 1 through 7.** Write the letter of the correct definition next to each term.

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|--------------------------|-----------------------------------------------------------------------|
| _____ 1. Analemma        | a) The place where a developable surface touches the globe            |
| _____ 2. Coriolis effect | b) A figure-eight diagram of the Sun's position across a year         |
| _____ 3. Dew point       | c) The energy needed to raise one gram of a material by 1.0°C         |
| _____ 4. Katabatic wind  | d) The apparent deflection of moving air caused by Earth's rotation   |
| _____ 5. Occluded front  | e) The region where solar radiation ionizes gas molecules             |
| _____ 6. Ionosphere      | f) Cold air sinking from a high plateau through gaps in the mountains |
| _____ 7. Specific heat   | g) The temperature at which saturated air condenses                   |
| _____ 8. Tangent point   | h) Where a cold front catches up to a warm front                      |

**27. Set 2 — Units 8 through 11.** Write the letter of the correct definition next to each term.

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|--------------------|------------------------------------------------------------------------------|
| _____ 1. Batholith | a) A sandstone containing more than 10% feldspar                             |
| _____ 2. Sill      | b) A non-foliated rock formed by contact metamorphism of fine-grained rock   |
| _____ 3. Arkose    | c) The color of a mineral's powder                                           |
| _____ 4. Breccia   | d) A pluton with an exposed surface area greater than 100 km <sup>2</sup>    |
| _____ 5. Evaporite | e) A rock formed as a body of water dries up and dissolved salts crystallize |
| _____ 6. Phyllite  | f) A tabular pluton that is concordant with existing layering                |
| _____ 7. Hornfels  | g) A foliated rock between slate and schist in grade, with a micaceous sheen |
| _____ 8. Streak    | h) A clastic rock dominated by angular clasts larger than 2 mm               |

**Part C · Short Answer**

6 questions × 3 points = 18 points

**28.** The scientific method is usually drawn as a fixed sequence of steps. Explain why real scientific inquiry rarely proceeds in that exact order.

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29. What is *redshift*, and what does it tell astronomers about the universe?

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30. State the principle of superposition, and describe one situation in which it would *not* hold true.

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31. Explain why Greenland appears roughly the same size as Africa on a Mercator map, when it is actually only about 7% of Africa's land area.

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32. Describe the path a rock takes through the rock cycle from magma to sedimentary rock. Name the processes involved.

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33. What is *foliation*, and what does its presence tell you about the conditions a metamorphic rock experienced?

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34. Follow a single grain of quartz through the rock cycle. It begins locked inside a granite batholith deep in the crust. Describe every step it takes to end up as part of a metamorphic rock. Name the processes at each stage and at least one specific rock type along the way.

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35. Earth's atmosphere is what makes the planet habitable. Explain at least three distinct ways it does so. For each one, name the specific layer, gas, or process responsible.

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